

Ila Rehman

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Summary

MCA graduate and self-taught AI engineer with a physics background, focused on shipping LLM-powered applications. Built a multi-tenant AI sales agent (LangGraph, FastAPI) and a production RAG API (FastAPI, LangChain, FAISS, GPT-4o), and currently build distributed, event-driven AWS microservices with LLMs integrated into production pipelines as a software engineer intern.

Experience

Manrke

April 2026 – Present

Software Engineer Intern

Remote

- Design and build distributed, event-driven microservices on AWS (Lambda, ECS, Step Functions, SQS/SNS, S3, DynamoDB, RDS) powering an end-to-end document-automation platform.
- Own systems across the full lifecycle: architecture, implementation, containerized deployment to ECR/Lambda across multiple AWS accounts, monitoring, and live production debugging.
- Integrate LLMs into production pipelines for extraction, classification, and natural-language generation, with guardrails, deterministic fallbacks, and cost/latency controls.
- Engineer for reliability at scale: idempotent processing, exactly-once semantics, distributed state coordination, and graceful failure handling across asynchronous services.
- Drive observability and operations: structured logging, correlation-ID tracing through CloudWatch, and alerting that surfaces and diagnoses issues quickly.

Projects

AI SDR (Davis) [\[code\]](#) [\[live demo\]](#) *Next.js, React, TypeScript, FastAPI, LangGraph, Groq, Supabase, Pinecone, Redis*

- Built an evidence-grounded AI sales-development agent that finds prospects, researches them, and drafts personalized outreach, where every prospect score cites a source URL, snippet, and observed-at timestamp.
- Split scoring into an LLM signal-extraction stage and a deterministic Python scorer with weighted ICP criteria, so identical inputs always produce identical, unit-testable scores; added an approval-first send pipeline (pending, approved, sent) with a 6-rule policy guard that makes duplicate sends impossible via a unique idempotency key.
- Shipped multi-tenant isolation across organizations and workspaces enforced by row-level-security policies over 45 Postgres tables, with Supabase auth, per-workspace Pinecone namespaces, and Braintrust tracing; deployed frontend on Vercel and backend on Railway.

AdaptQuiz API [\[code\]](#)

Python, FastAPI, LangChain, FAISS, GPT-4o, Docker, Railway

- Built and deployed a production RAG API that turns uploaded PDF/TXT study material into adaptive quizzes grounded in the source text, removing hallucinated questions.
- Built an LLM-as-judge grader (GPT-4o) scoring answers on a 3-criterion rubric (accuracy, completeness, terminology) with partial credit and per-question knowledge-gap tags, returned as structured JSON; served over 6 REST endpoints (OpenAPI/Swagger, Pydantic v2), containerized with Docker on Railway.

Technical Skills

Languages: Python, SQL, JavaScript / TypeScript

AI & LLM: LLM APIs (OpenAI GPT-4o, Groq, Anthropic Claude), RAG, LangChain, LangGraph, AI Agents, Prompt Engineering, Structured Outputs, Tool / Function Calling, LLM-as-Judge Evaluation, Embeddings (sentence-transformers), Semantic Search, FAISS, Pinecone

Backend: FastAPI, REST API Design, Pydantic, SQLAlchemy / Alembic, PostgreSQL, Supabase, DynamoDB, Redis, Event-Driven Microservices, Multi-Tenant Architecture, Row-Level Security, Next.js

Cloud & DevOps: AWS (Lambda, ECS, Step Functions, SQS/SNS, S3, DynamoDB, RDS, ECR, CloudWatch), Docker, Railway, Vercel, Git & GitHub, CI/CD

Tools & Foundations: pytest, Braintrust, OpenAPI / Swagger, NumPy, Pandas, ML Evaluation Metrics, Linear Algebra, Statistics

Education

Master of Computer Applications (MCA)

2022 – 2025

Indira Gandhi National Open University (IGNOU), New Delhi

Coursework: Software Engineering, Data Science, Cloud Computing, Artificial Intelligence & Machine Learning

B.Sc. (Hons) Physics with Mathematics

2017 – 2020

Jamia Millia Islamia, New Delhi

First Class Honours